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Review Article

Role of Biochar as a Feed Additive on Animal Performance, Digestibility, Micro-Biota Dynamics, and Reduction of Enteric Methane Production

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Biochar is a carbonaceous material resulting from the pyrolysis of biomass carried out in a rotary kiln with little oxygen. Biochar provides many advantages for animal nutrition when utilized as an ingredient in a blended product, including improved animal feed digestibility, reduced enteric methane production, and increased livestock growth performances for different animal species. For instance, the inclusion of biochar increases body weight gain for fatten bulls from 479 to 481 kg, dairy cows from 94.3 to 96.6 kg, layers from 1.2 to 1.3 kg, broilers from 1.8 to 2 kg, Pig from 36.5 to 40.5 kg, goat from 17.4 to 18.6 kg, and for Sheep from 22.1 to 25.2 kg. The inclusion of biochar feed additives in animal feed is important in satisfying the production performance of the animal. Feed additives are the most important source of feed supply to livestock production. Including biochar in animal feed results in greater feed conversion rates, digestion, body weight gain, growth rate, and microbiota production in the rumen. In addition to this, it can decrease the synthesis of intestinal methane production in different livestock breeds. In cattle, it can reduce from 30% to 40%, in goat from 0.033 to 0.02 mL/g, and in Poultry from 101.3 to 55.1 mL/g. In general, the importance of biochar as a feed additive, such as its direct effects on animal growth performance, feed digestibility, microbiota dynamics, and lowering enteric methane production in animal diets, is summarized in this review.

Keywords: biochar; digestibility; methane reduction; microbiota dynamics

1. Introduction

Livestock production is one of the major activities for human beings to generate varieties of products including milk, egg, and meat used for human food. It is also essential to the nutritional worth and economy of the world [1]. Therefore, improving animal feed must be considered to satisfy the economic and nutritional aspects of human beings [2]. People of the world should take into account both quantity and quality of feed to fulfill animal nutritional requirements and to increase animal production and productivity [3, 4].

Animal production and productivity are decreased due to metal toxicity and bioaccumulation by the food chain, which could result in harmful outcomes in animals, plants, and their consumers [5, 6]. Nickel (Ni), mercury, and cadmium (Cd) are among those toxic and highly water-soluble metals from contaminated soil, agricultural waste, grazing pastures, and vegetation [7–9]. Animals that are poisoned by mercury cannot produce liver, meat, or kidneys suitable for human consumption. Because cell toxicity and dysfunction can be caused by heavy metals through binding to protein sites and displacing the original metals from their native binding sites [9].

Therefore, inclusion of biochar is the best solution to meet the animal nutrient requirements and modulation of heavy metals. It can improve adsorption of organic compound nutrients like, fatty acids, urea, and amino acids.

Urea is adsorbed by biochar particles when utilized as bedding material. This preserves urea molecules from volatilizing and ammonifying that improves nutrient requirement of animals through retention and slow release of nitrogen, phosphate, and potassium [7, 10]. Similarly, biochar can immobilize heavy metals from contaminated feed and soil through surface adsorption (create binding sites with heavy metals) and ion exchange (reduces the mobility of heavy metals) [11, 12].

The history of biochar used for soil activities began about 7000 years ago [13]. It can be prepared from organic materials like wood, straw, manure, crop leftovers, and leaves heated from 350 to 1000°C in anaerobic conditions to partially pyrolyze [14, 15]. In the past 10 years, using and manufacturing biochar have been common and have become popular as a feed additive for animals and soil amendment for crop production [3].

Among the most common problems in livestock production industries include health issues, dietary nutrient deficiency-related issues (grass tetany, milk fever, hemorrhage, anemia, Keshan's disease, and goiter), and metabolic disorders such as carbohydrate metabolism (hypoglycemia, ruminal acidosis, and ruminal bloat), abnormal lipid metabolism (ketosis and milk fat depression in dairy cows, abnormal amino acid metabolism (edma, hyperhomocysteinemia, gout hyperammonemia, anemia) are the major one [16]. It has been reported that feeding biochar as a feed additive served as one of the best solutions to treat digestive disorders and metabolic abnormality [3, 10, 17]. Because biochar had the characteristics of toxin adsorption (bind toxins like dioxins, glyphosate, mycotoxins, and pesticides), improve nutrient intake and utilization, gut microbiota modulation through encouraging useful microbial populations and inhibits the growth of harmful bacteria, reduced methane production by preventing the activity of methanogen bacteria, and efficiency of nutrient retention leads to improve overall animal health and digestive disorder treatment [18, 19].

In the current scenarios, the absence of enough feed additives for livestock production, the increase in greenhouse gas (GHG) emission from ruminant animals, and the digestive or metabolic disorders due to malnutrition system for livestock production need alternative measurements. Of those available alternative measurements, previous studies have recommended that the inclusion of biochar in a feed has paramount importance. It can increase animal growth performance and health status [17, 20] and reduction of enteric methane production [21, 22]. However, besides the above importance, long-term feeding of biochar may have a risk. Because biochar had a potential to adsorb drug, essential feed compounds and shifts the microbiota dynamics in the digestive system of animals [23].

Therefore the objective of this review is to summarize existing studies on the role of biochar as a feed additive in animal performance, its importance for immunization of the digestive system, micro-biota dynamics, and reduction of enteric methane production. The review will summarize and indicate the maximum inclusion level of biochar as a feed additive for different group of animals.

2. Methods of Review

The topic was proposed based on its current issues and the importance of biochar for livestock production and methane emission mitigation. An understanding of the effect of biochar on animal performance, digestibility, micro-biota dynamics, and reduction of enteric methane production is very important. This review followed a systematic review method. Research articles on the use of biochar as feed additive were collected via using common search engines including Google Scholar, PubMed, and Web of Science with search terms such as biochar, feed additives, methane emission reduction, the effect of biochar on livestock performance, and methane emission (Figure 1). Then, after a screening of related and appropriate articles from different journal sites, articles were downloaded and stored in the EndNote library. Further screening was conducted to remove similar articles or duplicated articles.

2.1. Inclusion or Exclusion Criteria. The inclusion criteria for this systematic review were as follows: (1) a journal article published in English from a known publisher; (2) the experimental design should be fulfilled with the correct statistical rules; (3) the number and correctness of the experimental design and replicated material fit with the correct statistical standards or procedures; and (4) the experimental animal used.

2.2. Data Extraction and Analysis. From the selected articles, the author or sources, control (0% biochar), biochar inclusion level, source of biochar, animal species or host, microbiota dynamics, and effect on methane reduction (Tables 1–6) were extracted. Approximately 110 articles that were studied on the role of biochar as a feed additive on animal performance, digestibility, micro-biota dynamics, and reduction of enteric methane production were collected, but based on their title, abstract, and year of publication, only 100 of them were cited (Figure 1). Biochar inclusion level (% or g/kg of feed), microbiota dynamics (% or log10 cfu/g), and methane emission (% or ppm) were considered to be included in the parameters of comparison. All different functions like the exponential function were converted in a log function to calculate the average microbiota dynamics (Table 1). A simple meta-analysis was conducted to calculate the average and trend projections for the influence of biochar on all important variables (Table 7).

3. Year-Wise Publication, Trend Analysis, and Area of Research Focus in Biochar Application

As presented in Table 1, the data collected from the Web of Science revealed 22,620, 176,534, 63,620, and 30,835 for feed additive, improvement of rumen microbiota, animal feed digestibility, and enteric methane reduction, respectively. The overall published data were 293,609 on biochar applications for the livestock sector. This data were higher than the 6005 published data reported by Kumar et al. [29] in the application of biochar for the management of waste, reduction of climate change, soil quality improvement, energy production,

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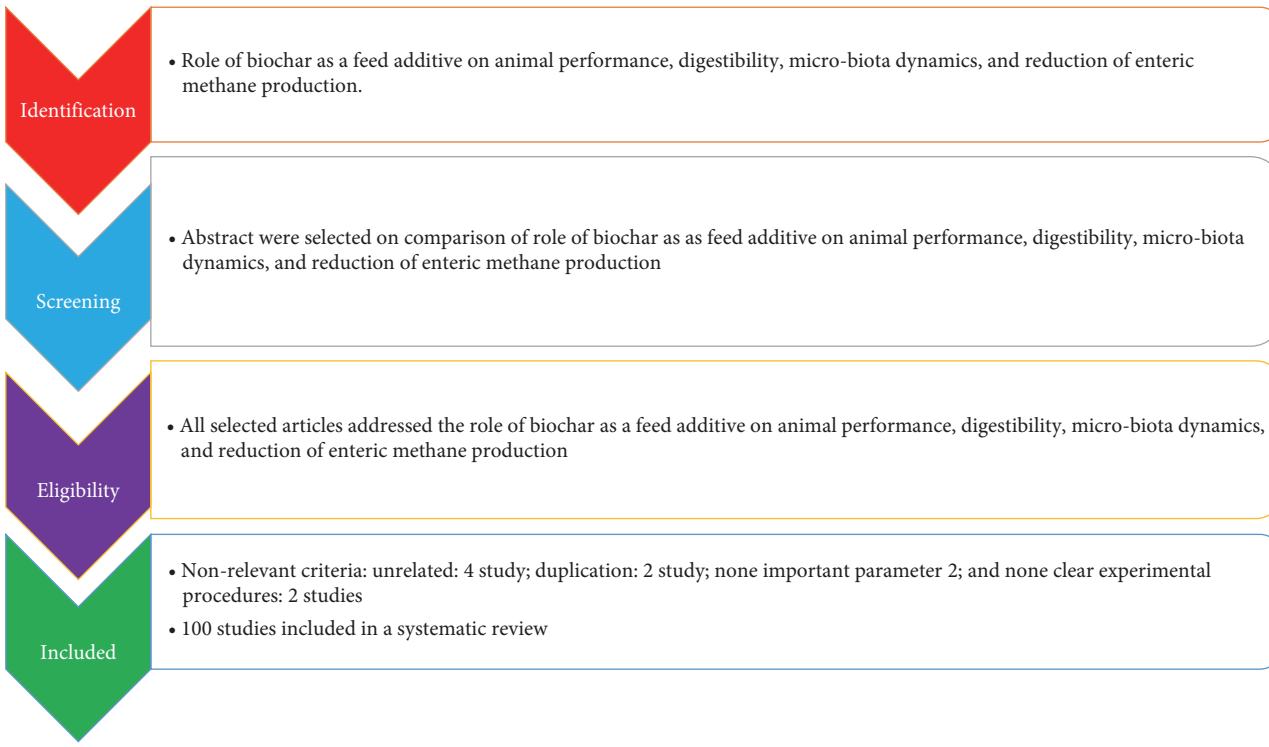


Figure 1: Selection of articles and evaluation process using the systematic review method.

Table 1: Search queries used for identifying publications on biochar application.

Application of biochar	Search query	Query link	Number of publications (extracted on11/02/2025)
1. Feed additive	Biochar "feed additive" OR Biochar "feed supplement"	https://www.webofscience.com/wos/woscc/summary/7634a92ba54c4675a57e99a17cc5ac23-0148986689/relevance/1	22,620
2. Improve rumen microbiota	Biochar "digestive microbiota" OR Biochar "rumen microbiota"	https://www.webofscience.com/wos/woscc/summary/8661284be2e3422ca8fa60d4f630966f0148a2c537/relevance/1	176,534
3. Improve digestibility	Biochar "improve digestibility"	https://www.webofscience.com/wos/woscc/summary/d4a77302c24645d081606397101d4dd3-0148a06882/relevance/5	63,620
4. Enteric methane reduction	Biochar "enteric methane reduction" OR Biochar "reduction of methane"	https://www.webofscience.com/wos/woscc/summary/8e11e74c24d4442b81a89ad2dc6e7d49-014899f92e/relevance/1	30,835
5. Overall	—	—	293,609

and remediation of contaminants from soil, water, and air. This difference might be the increase in research interest, data collection methods, or its wider importance in livestock sector.

The exponential trend analysis for year-wise publication for biochar application was presented in Figures 2 and 3. Based on this result, the application of biochar as a feed additive, improvement of rumen microbiota, animal feed digestibility, and enteric methane reduction in the livestock sector was increased exponentially. From these research topics, the improvement of rumen microbiota was the first followed by animal feed digestibility.

4. Use of Biochar as a Feed Supplement

While using biochar as an animal feed supplement, considering the regulation and feed certification under the European Food Safety Authority (EFSA) is mandatory. The mandate concerns feed safety, rules for organic livestock feeds, sustainability of organic production, labeling of organic products, and repealing restrictions [30]. The commission has checked and approved the biochar made from important plant parts, such as roots, leaves, wood, and feedstock free from any painting and organic and nonorganic contaminants, such as plastic [3].

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Table 2: Advancements in methodologies on the application of biochar.

Area of research Focus/Applications	Advancements in methodologies	Major findings	Sources
Livestock farming	Bedding, manure management to reduce ammonia discharges	Lower carbon, increase production, health and welfare of animals	[58]
Soil improvement	Precision biochar application based on soil type and crop needs	Increases crop yields ∼by 20% with about 10 t ha−1	[59]
Carbon sequestration	A key tool for large-scale carbon credit markets	Reduce greenhouse gas, enhances soil fertility	[60]
Waste management	Industrial, municipal waste and plastic pyrolysis	Waste management, climate change mitigation	[61]
Water treatment	Biochar-based nanocomposites for advanced filtration systems	Pollution removal, catalysis, modifying structure, composition	[62]
Biofuel production	Integration into bio-refinery systems for renewable energy production	Reducing greenhouse gas emission, cost-effectiveness	[63]
Construction & materials	Biochar-infused concrete, asphalt, and insulation	Improve bending ability, break hardiness cement mixtures, durability and flexural strength	[64]
Climate change mitigation	Focus on large-scale biochar projects for carbon offset credits	Improve bioenergy, soil fertility, and carbon stability	[65]
Microbial interactions	Improving microbial population to enhance soil health and nutrient cycling	Improve cumulative microbial respiration, decomposition, and metabolism of microbial necromass by other microorganisms	[66]
Industrial uses	Biochar as an additive in batteries, coatings, and composites	Filler for polymer, improved mechanical, electrical conductivity, thermal stability	[67]

Table 3: Maximum inclusion level of biochar and its effect on animals' performance.

Animal categories	Body Wt (Kg) without biochar	Body Wt (kg) With biochar	Maximum inclusion level	References
Cattle	138	143	1%	[69]
Fatten bull	479	481	0.80%	[70]
Dairy cow	94.3	96.6	0.60%	[33]
Average (kg)	237.1	240.2	0.8%	—
Broiler	1.824	1.776	3%	[71]
Broiler	1.46	1.465	3%	[72]
Layer	1.263	1.349	1%	[73]
Average (kg)	1.516	1.530	2%	—
Fingerlings	30.1	39.5	1%	[42]
Fingerlings	14.84	22.42	2%	[74]
Nile tilapia	22.3	39.5	10%	[42]
Average (gr)	22.41	33.81	4%	—
Pig	107	101	1%	[75]
Pig	29.8	31.6	2%	[74]
Average (kg)	68.4	66.3	1.5%	—
Goat	15.4	17.1	1%	[76]
Sheep	22.1	23.5	5%	[36]
Sheep	36.5	45.41	1.5%	[45]
Sheep	43.1	46.15	0.2%	[77]
Average (kg)	29.35	33.04	1.8%	—

Note: Kg = kilogram, gr = gram, Wt = Weight.

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Table 4: Inclusion of biochar to animal feed for growth performance (kg) or gram for fish and chicken animals.

Animals	Feed stock	BCF (day)	Biochar inclusion level (%, g and or kg/100) and results in Parentheses ()			Weight increase %	Blend	*p*-Value	References
Fatten bull	Pine	111	0% BC/DMI (479)	0.8% BC/DMI (481)	—	2	No	0.05	[70]
Dairy cow	Rice husks	98	0% BC/DMI (94.3)	0.6% BC/DMI (96.6)	6% Nitrate/DMI (94.5) 1.83% Urea/DMI (96.4)	2.3	No	0.69	[33]
Cattle	Rice husk	120	0% BC/DM (138)	0% RDB/DMI (133) 1% BC/DMI (143)	4% RDB/DMI (148)	7	GA	0.01	[123]
Broiler	Beechwood	35	0% BC/DM (1824^{a})	2% BC/DM ($1794^{a,b}$) 4% BC/DM ($1758^{a,b}$)	3% BCGA/DM ($1820^{a,b}$) 6% BCGA/DM (1701^{b})	4 reduced	GA	0.0059	[71]
Layer	Rice husk	56	0% FBio/DM (1263)	1% FBio/DM (1349)	0% LBio/DM (1305) 1% LBio/DM (1308)	86	No	0.22	[73]
Broiler	Rice husk	35	0 kg/100 kg of feed (1.46)	2 kg/100 kg feed (1.28) 4 kg100 kg feed (1.38)	2 kg/100 kg litter (1.51) 4 kg100 kg litter (1.42)	0.05	No	<0.05	[72]
Fingerlings	Active charcoal	56	0 g/kg diet (30.1^{a})	2 g/kg diet (34.1^{b}) 5 g/kg diet (36.7^{c})	10 g/kg diet (39.5^{d}) 20 g/kg diet (34.7^{b})	9.4	No	<0.05	[42]
Fingerlings	Parthenium, FM, PW, Vegetable, CW	90	0% BC/diet (14.84^{a} g)	2% Par (12.37^{c}) 2% FM/diet (19.5^{d}) 2% CW/diet (16.46^{b})	2% PW/diet (22.42^{e}), 2% Veg/diet (18.17^{f})	7.58	No	<0.05	[74]
Pig	Rice husk	28	0% /DM (107^{a})	4% RDB/DM (109^{a}) 1% BC/DM (101^{a})	4% RDB + 1% BC/DM (95.7^{a})	2	RDB	0.455	[32]
Pig	BLSO	91	0%/ diet (29.8)	1% BC/diet (28.8)	2% BC/diet (31.6)	2.2	No	0.665	[124]
Goat	Cassava root	120	0%/ DM (15.4)	30%WS/DM (17.5)	30%WS + 1%BC/DM (17.1)	2	WS	0.372	[76]
Sheep	Corncob	111	0 g/diet (22.1^{a})	1.5 g/day/diet (25.2^{b}) 3 g/day/diet (23.4^{b})	4.5 g/day/diet (23.5^{b})	3.1	No	<0.001	[36]
Sheep	WSB, PBB and CMB	90	0%/DM (36.5^{a})	PBB 1% /DM (43.10^{b}) WSB 1% /DM (45.3^{b})	1.5% CMB/DM (45.41^{b})	8.8	No	<0.001	[45]
Nile tilapia	Active charcoal	56	0% AC/kg diet ($14.5–30.1^{a}$) g	2% AC/kg diet (34.1^{c}) g 5% AC/kg diet 36.7^{b}) g	10% AC/kg diet (39.5^{d}) g 20% AC/kg diet (34.7^{c}) g	9.4	No	—	[42]
Sheep	Whole trees	55	Alf + Bar (43.4) g/d)	Alf +Bar + 2 g (kg) BC (46.15) g/d)	Alf + Bar or BC, Choice (43.4) g/d	0.05	Alf, Bar	0.334	[77]

Note: BC + GAS = biochar–glycerin–aluminosilicates mix, RDB, rice distillers' by-Product at 4% of diet DM, LBio, biochar in litter, FBio, biochar in feed, BC1 (beech, larch, spruce, and oak), BC2 (oak), BA + WS + BC = 70% *Bauhinia acuminata* + 30% water spinach + 1% biochar, BA + WS = 70% *B. acuminata* + 30% water spinach, BA + BC, *B. acuminata* ad libitum + 1% biochar, g/d = gram per day, Means having the same letter in the same column are not significantly different at $p < 0.05$.

Abbreviations: AC, active charcoal; BA, *B. acuminata* ad libitum; BCF, Biochar feeding (days); BLSO, beech, larch, spruce, oak, parthenium; CMB, chicken manure biochar; CW, corncob waste; FM, farmyard manure; Par, parthenium; PW, poultry waste, vegetable; Veg, vegetable.

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Table 5: Effect of biochar on microbiota dynamics.

Hosts invivo	Microorganism dynamics	Without biochar in log10 CFU/g	With biochar in log10 CFU/g	Maximum inclusion level	References
Pig	Coliform bacteria	3.56	2.25	0.3	[24]
Pig	Lactic acid bacteria	5	6.84	0.3	[24]
Pig	Salmonella	4.56	1.51	0.3	[24]
Holstein's calves	Cellulolytic bacteria	8.74	8.84	0.1	[28]
Beef Heifers	Protozoa	5.9	5.7	0.75	[23]
Holstein's calves	Protozoa	4.08	3.91	0.1	[28]
Sheep	Rumen protozoa	8.01	11.78	1.16	[45]
Layer	*E. coli*	6.15	5.90	0.1	[73]
Layer	Coliform	6.06	5.8	0.1	[73]
Average	—	5.8	5.83	0.36	—
Hosts invitro	**Microorganism dynamics**	**Without biochar %**	**With biochar %**	**Maximum inclusion level**	**—**
RUSITEC	Lachnospiraceae	17.4	16.8	0.02	[27]
RUSITEC	Prevotellaceae	13.7	12.43	0.02	[27]
RUSITEC	Ruminococcaceae	8.3	9.2	0.02	[27]
RUSITEC	Spirochaetaceae	6.7	6.9	0.02	[27]
RUSITEC	Rikenellaceae	4.8	6.1	0.02	[27]
Beef heifers	Entodinium	87.4	86.5	0.75	[23]
Beef heifers	Holotrichs	4.37	4	0.75	[28]
Chicken manure	Methanosarcina	71.18	73.9	0.05	[125]
Chicken manure	Methanosaeta	14.08	16.06	0.05	[125]
Chicken manure	Methanoculleus	5.29	7.88	4.90	[125]
Average	—	21.2	21.8	0.6	—

Abbreviations: CFU, colony-forming unit; g, gram.

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Table 6: Effect of biochar inclusion on microbiota dynamics (bacterial).

Hosts &	Bacterial dynamics	Feedstock	Biochar inclusion level (% or g) and results in parentheses ()				Blend	Types of trails	*p*-Value	References
Pig	Coliform bacteria log10 cfu/g	Bamboo	BD (3.56[a])	BD + 0.3% PC (0.98[b])	BD + 0.3% BC (2.25[b])	BD + 0.3% BV (1.10[b])	No	Invivo	<0.05	[24]
Pig	Lactic acid bacteria log10 cfu/g	Bamboo	BD (5.15[a])	BD + 0.3% PC (7.87[b])	BD + 0.3% BC (6.84[b])	BD + 0.3% BV (7.56[b])	No	Invivo	<0.05	[24]
Pig	Salmonella log10 cfu/g	Bamboo	BD (4.56[a])	BD + 0.3% PC (0.00)	BD + 0.3% BC (1.51[b])	BD + 0.3% BV (0.00)	No	Invivo	<0.05	[24]
Poultry	Proteobacteria log10 cfu/g	Woody green waste	Layer diet	Layer diet + 4% w/w biochar	Layer diet + 4% w/w bentonite	Layer diet + 4% w/w zeolite	No	Invivo	0.015	[25]
Poultry	Overall bacterial community	Woody green waste	Layer diet	Layer diet + 4% w/w biochar	Layer diet + 4% w/w bentonite	Layer diet + 4% w/w zeolite	No	Invivo	>0.05	[25]
Poultry	Campylobacterales	Woody green waste	Layer diet[a]	Layer diet + 4% w/w biochar[a]	Layer diet + 4% w/w bentonite[b]	Layer diet + 4% w/w zeolite[a]	No	Invivo	<0.05	[25]
RUSITEC	Bacterial	Hardwood	0 mg/day[a]	400 mg/day[a]	800 mg/day[a]	—	No	Invitro	>0.05	[26]
RUSITEC	Lachnospiraceae %	Stem wood chips	0 g/DM (17.4)	20 g/DM + $ZnCl_2$ (16.4)	20 g/DM + H_2SO_4 (17.1)	20 g /DM + HCl (16.9)	$ZnCl_2$, HCl H_2SO_4	Invitro	0.95	[27]
RUSITEC	Prevotellaceae %	Stem wood chips	0 g/DM (13.7)	20 g/DM + $ZnCl_2$ (11.9)	20 g/DM + H_2SO_4 (13.4)	20 g/DM + HCl (12.0)	$ZnCl_2$, HCl H_2SO_4	Invitro	0.86	[27]
RUSITEC	Ruminococcaceae%	Stem wood chips	0 g/DM (8.3)	20 g/DM + $ZnCl_2$ (9.0)	20 g/DM + H_2SO_4 (9.2)	20 g/DM + HCl (9.5)	$ZnCl_2$, HCl H_2SO_4	Invitro	0.72	[27]
RUSITEC	Spirochaetaceae%	Stem wood chips	0 g/DM (6.7)	20 g/DM + $ZnCl_2$ (7.3)	20 g/DM + H_2SO_4 (6.2)	20 g/DM + HCl (7.3)	$ZnCl_2$, HCl H_2SO_4	Invitro	0.41	[27]
RUSITEC	Rikenellaceae%	Stem wood chips	0 g/DM (4.8[b])	20 g/DM + $ZnCl_2$ (6.1[a])	20 g/DM + H_2SO_4 (6.5[a])	20 g/DM + HCl (5.7[a])	$ZnCl_2$, HCl H_2SO_4	Invitro	0.05	[27]
Holstein's calves	Cellulolytic bacteria (Log10/mL)	Pomegranate and plum woods	Diet (8.74[a])	Diet + biochar (8.84[b])	Diet + lactobacilli-biochar (8.86[b])	Diet + yeast-biochar (8.93[d])	LB, yeast	Invitro	0.007	[28]

Note: Means having the same letter in the same column are not significantly different at $p<0.05$.

Abbreviations: BC, biochar; BD, basal diet; BV, bamboo vinegar; cfu/g, colony-forming unit; LB, lactobacilli; RUSITEC, rumen simulation technique; ZT, bentonite; ZT, zeolite.

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Table 7: Effect of biochar inclusion on microbiota dynamics (continued…….)

Host	Microbial dynamics	Feedstock	Biochar inclusion levels and its effect in Parentheses ()			Blend	Types of trial	*p*-Value	References
RUSITEC	Fungal	Hardwood	0 mg/day	400 mg/day	800 mg/day	No	In vitro	>0.05	[26]
RUSITEC	Archaeal	Hardwood	0 mg/day	400 mg/day	800 mg/day	No	In vitro	>0.05	[26]
Beef Heifers	Protozoa $n \times 10^5$	Yellow pine	0%/DM (8.37^a)	0.5% (6.10^b)	1% DM/ (5.68^b)	No	In vivo	<0.01	[23]
Holstein calves	Total protozoa (Log10/mL)	Plum woods	0 (4.08^a)	biochar (4.04^b)	Lactobacilli + biochar (3.78^c)	LB	In vitro	0.002	[28]
Beef heifers	Entodinium %	Yellow pine	0%/DM (87.4^b)	0.5% (87.1^b)	1%/DM (85.8^b)	No	In vivo	0.14	[23]
Beef heifers	Holotrichs %	Yellow pine	0%/DM (4.37^b)	0.5% (3.56^b)	1% /DM (4.45^b)	No	In vivo	0.21	[23]
Pig manure	Fungal %	Bamboo	0% Biochar	10% CSB	10% Bambo Biochar	No	In vitro	<0.05	[68]
Pig manure	Ascomycota %	Bamboo	0% Biochar	10% CSB	10% Bambo Biochar	No	In vitro	<0.05	[68]
Pig manure	Basidiomycota %	Bamboo	0% Biochar	10% CSB	10% Bambo Biochar	No	In vitro	<0.05	[68]
Pig manure	Mucoromycota %	Bamboo	0% Biochar	10% CSB	10% Bambo Biochar	No	Invitro	<0.05	[68]
Chicken manure	Methanosarcina %	Orchard waste wood	Inc + CM (52.5–89.86)		Inc + CM + 4.9% BC (58.2–89.6)	Inc	Invitro	—	[125]
Chicken manure	Methanosaeta %	Orchard waste wood	Inc + CM (8.16–20.0)		Inc + CM + 4.9% BC (8.12–24.0)	Inc	Invitro	—	[125]
Chicken manure	Methanoculleus%	Orchard waste wood	Inc + CM (2.15–8.39)		Inc + CM + 4.9% BC (1.83–13.92)	Inc	Invitro	—	[125]
Sheep	Rumen protozoa (log10/ [cfu/mL])	WSB, PBB and CMB	0% (8.01), WSB 1% (11.84)		PBB 1% (10.67, CMB 1.5% (12.83)	No	Invivo	0.13	[45]
Layer	*E. coli* (CFU/g)	Rice husk	FBio 0% (1.7×10^6) FBio 1% (8.2×10^5)		LBio 0% (1.3×10^6) LBio 1% (1.2×10^6)	No	Invivo	—	[73]
Layer	Coliform (CFU/g)	Rice husk	FBio 0% (1.4×10^6) FBio 1% (5.1×10^5)		LBio 0% (9.6×10^5) LBio 1% (9.1×10^5)	No	Invivo	—	[73]

Note: Means having the same letter in the same column are not significantly different at $p < 0.05$, LB = Lactobacilli (2 g/d per animal), Yeast (2 g/d per animal), Biochar (1% of diet) [28].
Abbreviations: BB, bamboo biochar; CMB, chicken manure biochar (1.5%); CSB, coconut shell; FBio, biochar in feed; Inc +CM, inoculums + chicken manure feed +biochar; LBio, biochar in litter; PBB, pistachio by-product biochar (1%);RDB, Rice Distillers' By-Product at 4% of diet DM; RUSITEC, rumen simulation techn; WSB, walnut shell biochar 1%.

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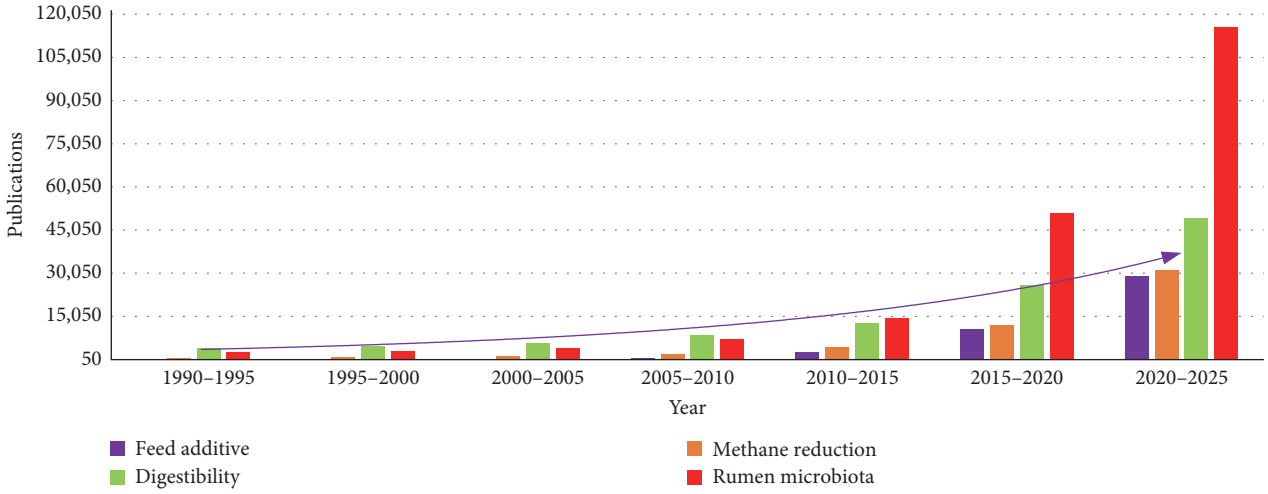


Figure 2: Number of published research articles from 1990 to 2025 per research topic.

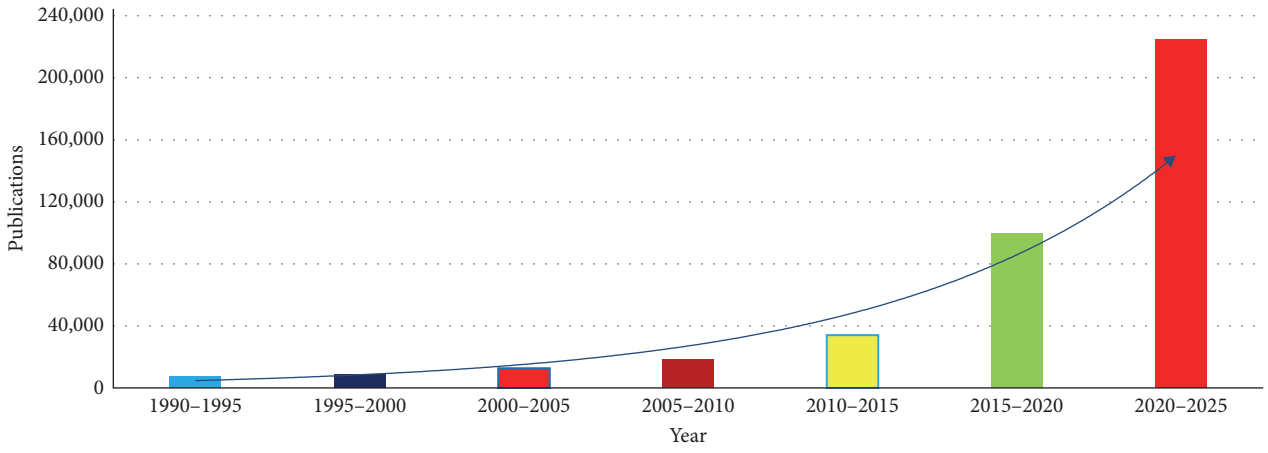


Figure 3: Overall published articles (1990–2025) for biochar applications in livestock.

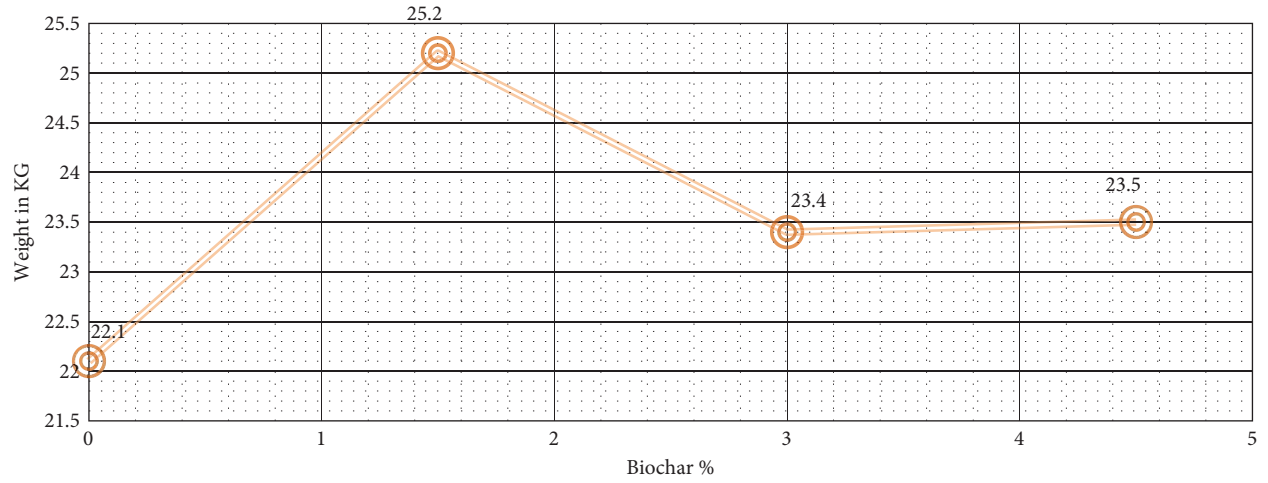


Figure 4: The effect of biochar on body weight gain of sheep.

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5. Effects of Biochar Inclusion to Feed on Animal Performance

In livestock farming, supplementing feed with feed additives is a widespread method to improve animal immunity, protein intake, growth efficiency, and livestock production [3]. Feeding a regular dosage of charcoal is widely used to improve animal health and growth performance [3]. Different studies have revealed that adding biochar to feed can help animals grow weight and improve appetite to provide more food [31, 32].

5.1. Impact of Biochar on Cattle Performance. Based on [33] findings, the weight of local yellow cattle was increased by 25% when 0.6% of rice husk-based biochar was added to the feed. In addition to this, Le Thi Thuy Hang, Ba and Van Dung [34] noted that the body weight of cattle was increased by 43% when rice husk-based biochar was added to urea-molasses blocks, and the feed conversion ratio (FCR) dramatically decreased from 16.4 to 10.7%.

5.2. Impact of Biochar on Small Ruminate Performance. The inclusion of biochar in the feed for small ruminate has a significant impact on growth performance. When compared to control groups, goats' weight gain was significantly affected by diets supplemented with 1.1% of biochar [35]. As presented in Figure 4, there was an increase in body weight gain by up to 1.5% inclusion of biochar in the sheep experiment [36]. However, when the inclusion level of biochar is greater than 1.5%, there was a reduction of body weight gain in sheep.

5.3. Impact of Biochar on Pig Performance. The results on pigs conducted by Sivilai and Preston [32] revealed that a positive impact on weight gain contained rice husk-based biochar and it was increased by 20.1% when 1% biochar was added to the diet and by 22.9% when it was coupled with rice distillers' by-products. Man et al. [3] conclude that the daily meal can be supplemented with 0.3%–3% biochar to improve pig growth performance and immunity.

5.4. Impact of Biochar on Poultry Performance. Feeding biochar as a feed additive for poultry also increases production performance, growth rate, egg production, and weight gain. Prasai et al. [25] noted that feeding wood-based biochar to layer chickens produced heavier eggs ($60.6 + 1.14$ g) compared to the control group. Dim et al. [37] stated that the inclusion of biochar in the feed had a substantial impact on broiler bird performance during the finisher phase, as seen by variations in final body weight, average daily weight gain (ADWG), and FCR between treatments. Birds given 6% biochar per kg had considerably higher final body weight, ADWG, and FCR than other treatment groups ($p < 0.05$). However, Dim et al. [37] reported that final body weight, ADWG, average daily feed intake, and FCR at the beginning phase revealed no significant differences ($p > 0.05$) between treatments. This is because young birds' digestive tracts are typically not fully matured and are, therefore, unable to properly absorb nutrition [38]. Prasai et al.[25] also stated that all treatment groups' laying performance had improved, and all of the differences were statistically significant ($p < 0.001$). When biochar is added to the diet for layers, the average egg mass, shell resistance to crushing, shell thickness, smell, white texture, yolk texture, and yolk color are all statistically significant ($p < 0.001$) for egg quality characteristics. There was variability on poultry performance from different studies using biochar as feed additive. This variability was due to difference in biochar properties (source of feed Stock), experimental conditions, duration of biochar feeding, breed (broiler and layers), biochar inclusion level, diets, and blended feed types.

5.5. Impact of Biochar on Fish Production. A different study revealed that adding biochar to fish feed has a prominent result [39]. According to Khaki [40], the largest brown trout (*Salmo trutta*) received biochar-supplemented meals 0.2 and 0.3 g per kg diet was higher weight gain. Growth rates were increased when 36% of biochar was fed to the striped catfish (*Pangasius hypophthalmus*) that biochar made from 1% of rice husks [41]. The application of active charcoal supplementation in the experiment of *Nile tilapia*, Oreochromis revealed it can improve fish growth performance and feed intake and the optimum active charcoal level was added to be 7.0 g/kg diet [42]. Generally, Figure 5 revealed that feeding biochar can increase animal growth performance per livestock species.

As Summerside here in Table 1, the maximum inclusion level (0.8%) for cattle can improve from 237.1 to 240.2 kg of body weight gain. Similarly, body weight was increased from 22.41 to 33.81 kg for fish under maximum inclusion level of 4% biochar and for shoat from 25.35 to 33.04 kg in maximum inclusion level (1.8%) of biochar in the feed. However, there was slight similarity in average result of poultry. In conclusion, our review found that biochar had a positive effect on animal body weight gain to the amount indicated by the maximum inclusion levels below.

5.6. Impact of Biochar on the Metabolic Activity of Ruminates. Metabolic activity and weight gain of ruminates can be improved via the inclusion of biochar. It has a positive impact on increasing nutrient uptake and improves the overall feed efficiency and the feed-to-weight ratio for livestock. The inclusion of biochar in ruminant animal diets has increased rumen fermentation and growth performance of beef cattle [33].

The high surface area and porous structure of biochar are postulated to promote biofilm formation, increasing microbial growth efficiency [43]. Recently, metabolism trials have found that rumen fermentation and microbial nitrogen (N) flow were not altered by the inclusion of biochar in the diets of beef heifers [23, 44]. In comparison to the control groups from the ram experiment conducted by Mirheidari, Torbatinejad [45] revealed that the addition of walnut shell biochar enhanced ($p < 0.01$) the total tract digestibility of dry matter (DM), organic matter (OM), crude protein (CP), and neutral detergent fibre (NDF). Lambs given either walnut shell biochar or pistachio by-product biochar had greater ruminal ammonia-N (NH3-N) levels than controls (p 0.05).

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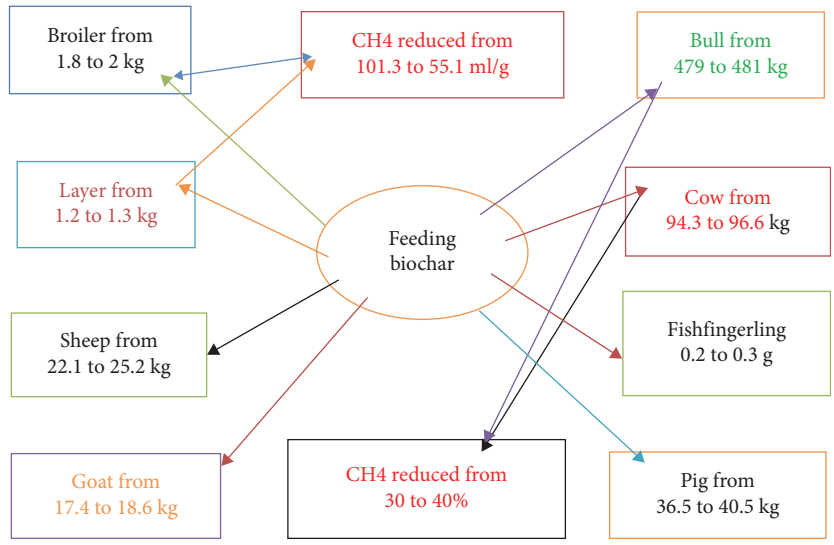


FIGURE 5: Conceptual diagram on the role of feeding biochar on animal growth performance production.

6. Biochar for Enhancement of Digestive Microbiota

Animal stomachs are more intricate ecosystems with a variety of microbiota. This microbiota exists in the rumen-reticulum, gastrointestinal tract, and cecum in a symbiotic relationship with the host [46]. They degrade any feeds consumed by the host, produce various biochemical substances, and play various roles in the production of volatile fatty acids (VFA), which are used as a source of energy in glycolysis [47]. Creating a solid surface region for microorganisms might effectively transfer substrate, increase the efficiency of adenosine triphosphate (ATP) synthesis, and biochar might promote the formation of specific rumen microbial populations [41].

Silage mixtures containing 8.8% and 16.6% of biochar led to increments in total VFA production, which suggests increased microbial fermentation with an increasing quantity of biochar [48]. This is because biochar can increase the number of lactic acid bacteria, restrict butyric acid production, and limit mycotoxin production.

According to Amean and Shujaa [49], Awassi lambs were fed varying percent of biochar supplements. Average daily gain (ADG), final body weight gain, and FCR were significant ($p < 0.05$) for ruminant animals due to an increase in bacterial population [33]. They stated that continuous increasing the percent of biochar content in animal feed, and performance may lower due to decreasing microbial productivity and higher urea or nitrogen absorption in the biochar's pores, which renders these nutrients inaccessible to microbial protein synthesis.

6.1. Microbiota Dynamics in Normal Cattle (bovine) Rumen. Scholars have documented that a single milliliter of rumen fluid from normal cattle contains ~10^{11} (99%) bacteria, 10^9 (0.99%) archaea, 10^6 (0.00099%) protozoa, and 10^6 (0.00099%) fungi, respectively. Rumen microbes fall into two main categories: those that digest fibre, preferring a pH above 6.2, and those that digest starch, with a preferred pH range between 5.4 and 7.0. These microorganisms exhibit sensitivity to the rumen environment and undergo fluctuations influenced by dietary changes. The consumption of a high-starch diet can induce a shift in rumen pH, leading to alterations in microbe populations [50]. Based on the average result presented in Figure 6, Bacteroidetes (27%), Prevotellaceae (21%), and Firmicutes (19%) were the most dominant bacterial population in the rumen of cows.

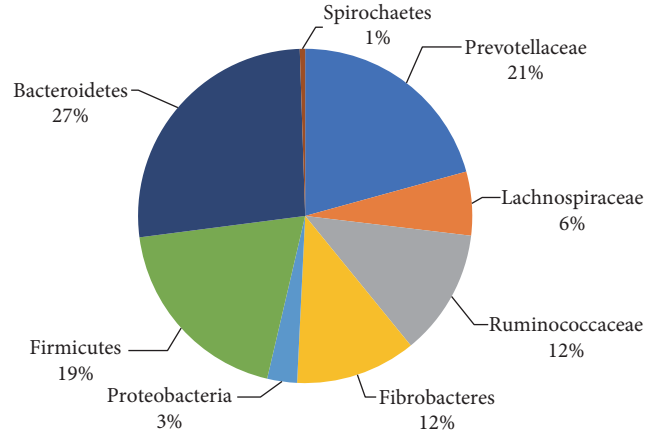


FIGURE 6: The major bacterial population in the rumen of a healthy cow [51–53].

7. Effect of Biochar Inclusion on Microbiota Dynamics

The exploration of microbiota dynamics in the rumen of animals involves investigating the intricate interactions within the distinctive digestive system of ruminants. Ruminant animals, such as cows, sheep, goats, and deer, exhibit remarkable microbial diversity within their rumen. In this intricate ecosystem, diverse microorganisms engage in interactions with one another [54, 55].

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Various microorganisms present in the rumen, including Bacteria, Protozoa, Fungi, Archaea, and Viruses, contribute significantly to nutrient processing. Their essential roles involve the conversion of plant fiber into short-chain fatty acids and other metabolites. This transformative process is crucial for the production of meat and milk, directly influencing the quality of feed. Furthermore, rumen microbes play a vital role in detoxification and the metabolism of xenobiotics, such as aromatic compounds and secondary plant metabolites [56, 57].

7.1. Effect of Biochar on Microbiota Dynamic. According to Prasai et al. [25], the inclusion of biochar at 4% prepared from woody green waste decreased the microbiota dynamics of proteobacteria in the cloaca of the laying hens ($p = 0.015$). However, the inclusion of biochar at 0.3% on Pig feed increased microbiota dynamics of *Lactobacillus*, anaerobic bacteria, and lactic acid bacteria. However, Chu et al. [24] reported that there was a reduction of Coliform bacteria, lactic acid, and *Salmonella* bacterial populations using biochar at 3% of inclusion levels. The population of Proteobacteria bacteria was not affected by biochar inclusion; however, it affected the Campylobacterales population which was conducted on Poultry [25]. On the other hand, the bacterial population of Lachnospiraceae, Prevotellaceae, Ruminococcaceae, Spirochaetaceae, and Rikenellaceae was not affected by biochar inclusion [27]; however, Cellulolytic bacteria was increased [28] (Table 2). Teoh et al. [26] also reported that the inclusion of 400 and 800 mg of biochar has no significant effect on fungal and archaeal population dynamics (*Vishniacozyma victoriae* and *Sporobolomyces ruberrimus* structure ($p > 0.05$. However, the inclusion of biochar has an effect on Protozoa dynamics in beef heifers [23]. Similarly, biochar affects fungi, Ascomycota, Basidiomycota, and Mucoromycota [68]. In general, as illustrated in Table 3, the overall average result for Invivo fermentation for all microorganisms was 5.8 and 5.83 in log10 CFU/g, respectively. On the other side, the overall result for invitro fermentation without biochar and with biochar was 21.2% and 21.8%, respectively.

7.2. The Negative Effect of Biochar on Gut Microbiota

1. Disruption of Microbial Balance: Biochar may disrupt the natural balance of gut microbiota, leading to dysbiosis. This can result in a decrease in microbial diversity and the imbalance of beneficial and harmful bacteria [78].
2. Promotion of Pathogenic Bacteria: Certain components in biochar might promote the growth of pathogenic bacteria, potentially leading to gastrointestinal issues and infections. Biochar can adsorb and release nutrients in a manner that may favor the growth of pathogenic bacteria. For instance, if biochar releases nutrients more slowly, it can create a steady supply that pathogens can exploit [79, 80].
3. Suppression of Beneficial Microbes: Biochar could suppress beneficial microbes, such as those involved in nutrient absorption and immune function, negatively impacting gut health. Biochar can affect the pH of the soil or gut environment and beneficial microbes often have specific pH ranges for optimal growth and shifts in pH can suppress their activity [80].
4. Introduction of Contaminants: Biochar can contain contaminants like heavy metals or organic pollutants depending on its source and production process. These contaminants can harm gut microorganisms, leading to toxicity and adverse health effects.
5. Alteration of Microbial Metabolism: The introduction of biochar might alter the metabolic activities of gut microorganisms, potentially disrupting normal metabolic processes and leading to negative health outcomes [81].
6. Physical Damage to Microbial Cells: The abrasive nature of biochar particles can physically damage microbial cells, which might reduce microbial populations and negatively affect gut health [82].
7. Impact on Microbial Communication: Biochar may interfere with microbial signaling pathways, affecting communication between gut microbes and disrupting their cooperative functions [83]. Additionally [84] reported that wood biochars interrupt communication within a growing multicellular system that is made up of sender cells which synthesize acyl-homoserine lactone. Biochar inhibition of acyl-homoserine lactone-mediated cell–cell communication varied, with the biochar prepared at 700°C.

8. Biochar Supplement for Enhancing Digestibility

Leng, Inthapanya, and Preston [85] illustrated that animals fed with biochar as a feed additive has improved feed digestibility and digestion rates. Similarly, Silivong and Preston [35] and Winders et al. [86] investigated that ruminants can digest feed improved by biochar in their diets [87] also confirmed a positive response in dry matter digestibility (DMD) using pine-sourced biochar. In addition to this [88], reported that biochar addition of the *in vivo* method improves the nutrition content and digestibility of mycotoxin-containing silages. However, Teoh et al. [26] noted that biochar is entirely inorganic matter and is not digested by the rumen microbiota. There was no effect on DMD with biochar supplementation in the study. However, there is evidence suggesting that biochar does not affect digestibility [89, 90].

9. Biochar Supplement for Enteric Methane Reduction

General information on biochar's contribution to the reduction of methane emission in different livestock species and from in vitro fermentation systems is presented in Table 8. Due to herbivorous animals' significant contribution to the production of enteric methane, altering rumen fermentation to reduce this greenhouse gas should be given a high priority in the research field [33]. As fermentation duration and methane output increase, relative biochar addition can

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Table 8: Effect of biochar on reduction of methane emission in different livestock species and from in vitro fermentation system.

Experimental test	Without biochar CH_4 gram/day	With biochar CH_4 gram/day	Maximum inclusion level (%)	References
Cattle	0.009	1.5	0.5	[91]
Goat	0.98	0.67	0.36	[34]
Goat	0.3	0.02	1.29	[34]
Growing steers	109	92.2	0.8	[44]
Finishing steers	141	122	3	[44]
Beef heifers	225.1	215.5	1	[23]
Yellow cattle	0.09	0.08	0.6	[33]
Dairy cows	322	348	4	[92]
RUSITEC	0.06	0.05	7	[27]
RUSITEC	0.0358	0.04	7.2	[26]
In vitro rumen	26.5	25.3	8.1	[93]
Average	75.01	73.21	3.1	—

reduce methane (CH_4) synthesis [85]. Although there is little study on the use of biochar in animal diets, several sources indicated that a decrease in methane (CH_4) gas production without any decline in digestibility [94].

Biochar is already used as a beneficial feed ingredient and has a market, and there is *in vivo* proof that it can reduce greenhouse gas emissions [17]. The average result for methane production from several researchers under in vitro and in vivo experiments was 73.21 g/day with a maximum inclusion level of 3.1% biochar (Table 8). Similarly, the result concluded by Winders et al. [44] in growing steers and finishing cattle, biochar given at 8 g/kg DM decreased methane (CH_4) generation by 9.5 and 18.4%, respectively. An experiment conducted by Cabeza et al. [95] also revealed that, compared to the control group, adding biochar decreased the production of methane (CH4) and total gas. Sirjani, Zahedifar, and Rouzbehan [28] also reported that there was a significant difference ($p < 0.05$) in methane gas production between control (21.6%), Diet + biochar (14.5%), Diet + lactobacilli-biochar (13.4%), diet + yeast-biochar (13.2%), and it was revealed that there was the reduction of methane gas production via feeding of biochar as a feed additive.

However Terler et al. [92] was reported that Supplementing with a 4% maximum inclusion level of biochar on dairy cows' feed will increase methane production from 322 to 348 grams/day (Table 8). More research on methane (CH_4) and hydrogen gas production *in vivo* models is needed to justify biochar's potential as a methane mitigation technique properly.

9.1. How Does the Addition of Biochar Reduce Enteric Methane Production? The addition of biochar to ruminant diets is considered a potential strategy to reduce enteric methane production. Enteric methane is produced during the digestion process in the stomachs of ruminant animals, such as cattle, sheep, and goats [96]. As presented in Table 9, the addition of biochar at 1.29% decreased methane production from 32.4 to 15.7 ppm [34], at 0.8% decreased from 109 to 92.2 g/day [44], at 3% decrease from 141 to 122 g/day [44], and at a 1% decrease from 225.1 to 215.5 g/day[23 for goat, growing and finisher Steers and Beef Heifers, respectively. Methane is a powerful greenhouse gas, and its reduction is important for mitigating climate change [97].

Here's how the addition of biochar may help in this regard:

1. Adsorption of Methane: Biochar has a porous structure with a large surface area. When mixed in the feed of ruminants, it can adsorb methane gas within its pores. This decreases the volume of methane released into the atmosphere during the digestion process [98, 99]. Biochar also has electron-mediating characteristics in biological redox reactions [100].
2. Microbial Interactions: Biochar can influence the microbiota dynamics in the digestive system of ruminants. Methanogenic archaea are microorganisms responsible for methane synthesis in the rumen. The inclusion of biochar may affect the microbial community, possibly decreasing the presence or activity of methanogenic archaea, leading to reduced methane release [101, 102].
3. Improved Digestive Efficiency: Biochar has the benefit of enhancing the overall digestive efficiency of ruminants. By improving nutrient consumption and fermentation processes in the rumen, biochar is used to a reduction in the fermentation of feed leading to, the mitigation of methane production [103, 104].
4. Changes in Fermentation Pathways: Biochar can adjust the fermentation pathways in the rumen. It may improve the synthesis of acetate and propionate, which are less methane-producing VFA compared to butyrate. This shift in fermentation pathways can result in lower methane emissions [43].
5. Altering pH Levels: Biochar can influence the pH levels in the rumen. When included in the rumen, biochar affects pH levels due to its buffering capacity. This means it can stabilize pH by absorbing or releasing hydrogen ions (H^+) as desirable.

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Table 9: Contribution of biochar to methane reduction

Animals	Feedstock	Biochar inclusion level (% or g) and results in parentheses ()			Blend	Types of trial	*p*-Value	References
Cattle	Powdered activated carbon	0% (9.14) ppm	—	0.5% (1500) ppm	No	Invivo	0.001	[91]
Goat	Rice husks	0% (982), 0.36% (669) ppm	0.85% (686) ppm	1.29% (709) ppm	No	Invivo	0.001	[34]
Goat	Rice husks	0% (32.4) ppm	0.85% (16.3) ppm 0.36% (18.2) ppm	1.29% (15.7) ppm	No	Invivo	0.001	[34]
Growing Steers	Whole pine trees	0% (109) g/d	0.8% (92.2) g/d)	3% (100) g/d	No	Invivo	0.68	[44]
Finishing Steers	Whole pine trees	0% (141) g/d	0.8% (128) g/d	3% (122) g/d	No	Invivo	0.39	[44]
Beef Heifers	Yellow pine	0% (225.1) g/d	0.5% (216.5) g/d	1% (215.5) g/d	No	Invivo	0.43	[23]
Yellow Cattle	Gasifier stove	0% (85.7)	0.6% (76.2)	—	No	Invivo	0.066	[33]
Dairy Cows	Pure ash wood	0 (322) g/day	200 (348) g/day	200 BC + 90 U (371) g/day	Urea	Invivo	0.210	[92]
RUSITEC	Stem wood chips	0 g (59.9) mg/day	20 g/DM + H_2SO_4 (64.5) mg/day	20 g/DM + HCl (67.6) mg/day 20 g /DM + $ZnCl_2$ (53.8) mg/day	H_2SO_4, HCl $ZnCl_2$	Invitro	0.23	[27]
RUSITEC	Hardwoods	0% (35.8^a) mg/day	3.6% (39.6^b) mg/day	7.2% (35.5^c) mg/day	No	Invitro	0.10	[26]
In Vitro Rumen	Pine wood chips and corn stover	0 g/kg (26.5) mL/g	81 g/kg CS (27.7) mL/g 81 g/kg pine (25.3) mL/g	186 g/kg CS (27.4) mL/g 186 g/kg pine (27.8) mL/g	No	Invitro	0.619	[93]

Note: Methane in % of total gas at (Control = 21.6^a), (Diet + biochar = 14.5^b), (Diet + lactobacilli-biochar = 13.4^c), (Diet + yeast-biochar = 13.2^c) *p*-value = 0.018 [28].
Abbreviations: BC + U, biochar + urea; CS, corn stover; g, gram; Mcal, megacalorie; mL, milliliter; ppm, parts per million; RUSITEC, rumen simulation technique.

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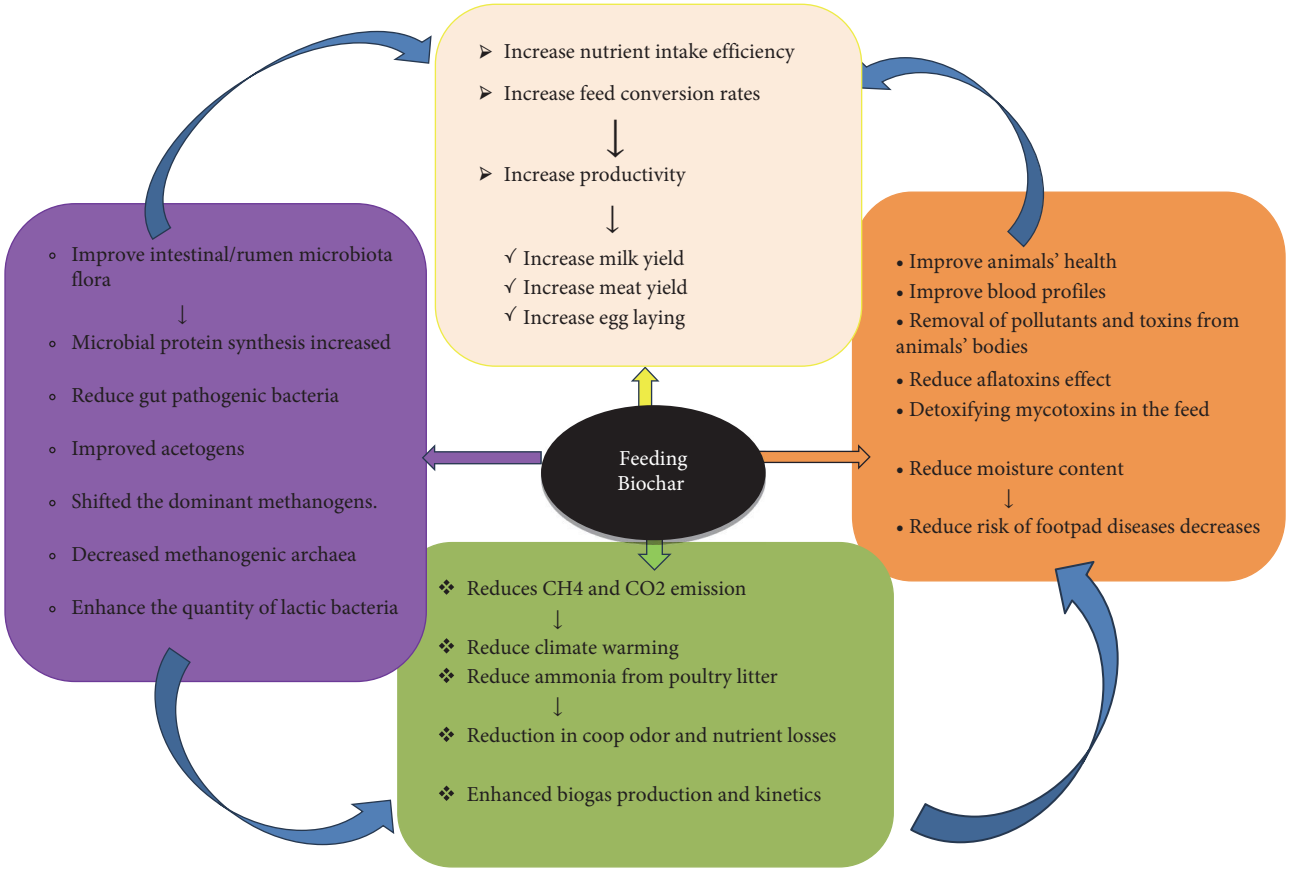


FIGURE 7: Conceptual framework on the role of biochar on livestock production, microbiota dynamics, and enteric methane reduction [3, 17, 45, 106–109].

Methane synthesis is sensitive to pH, and by keeping optimal pH states, biochar can potentially decrease the activity of methane-producing microorganisms [26, 105].

Generally, as the conceptual framework presented in Figure 7 revealed, biochar is important for livestock production, microbiota dynamics improvement, and enteric methane reduction.

9.2. Adsorption and Redox Activity of Biochar in rumen. The intricate microbial ecosystem found in the rumen is a key component of biochar's redox process in ruminant digestion. Cattle and other ruminant animals have a special digestive system that includes multiple chambers in the stomach. This stomach is called the rumen, and it is here that microbial activity occurs during fermentation to break down complex plant components [110]. Electron transfer reactions inside the rumen environment can be facilitated by the activity of biochar, which serves as an electron shuttle. This redox activity may have an impact on the rumen's total redox potential, which could alter microbial metabolism [111].

Biochar improves adsorption and redox processes by enhancing its overall digestibility for animal application and productivity [112]. Direct interactions between NO^{-}_{2} and the ruminal microbiota might suppress methanogenesis [113]. Both the pH and redox potential of biochar directly influences its electrochemical characteristics. Due to its redox and adsorption potential greatly impacts rumen biochemical processes, electrical conductivity, pseudo capacitance, and double-layer capacitance. Its pseudo capacitance can operate as a hydrogen sink by absorbing hydrogen generated by protozoa keeping it out of the way on the rumen environment and producing methane [114].

9.2.1. Adsorption Mechanism. Biochar possesses a high surface area due to its porous structure. This provides ample sites for the adsorption of various compounds, including organic molecules, nutrients, and contaminants [115, 116]. In addition, it contains functional groups on its surface, such as hydroxyl (—OH), carboxyl (—COOH), and phenolic groups. These functional groups enhance the adsorption of substances through electrostatic interactions, hydrogen bonding, and Van der Waals forces [117]. Biochar can also adsorb and retain nutrients such as nitrogen, phosphorus, and potassium. This nutrient retention capacity contributes to improved nutrient availability and enhances the digestibility of nutrients [110].

9.2.2. Stimulation of Microbial Activity and Methanogenesis Regulation. The redox-active properties of biochar have the potential to enhance rumen microbial activity. For an animal to more readily digest complex carbohydrates, microbes are essential in converting them into simpler forms like short-chain fatty acids [112].

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Table 10: Evaluation of cost-effectiveness of using biochar as a feed additive in a large-scale livestock production.

Animals	Weight gain in kg	Price Meat/kg	Total income	Price biochar/kg	Biochar used in kg	Total price/biochar	Reference
Cattle	3	$ 5.30	$ 15.90	$ 0.69	0.02	$ 0.01	[33, 69, 70]
Nile Tilapia	0.02	$ 5.30	$ 0.11	$ 0.69	0.04	$ 0.03	[42]
Pig	1.8	$ 5.30	$ 9.54	$ 0.69	0.03	$ 0.02	[74, 75]
Goat	2	$ 5.30	$ 10.60	$ 0.69	1	$ 0.69	[76]
Sheep	13	$ 5.30	$ 68.90	$ 0.69	0.007	$ 0.005	[36, 45, 77]
Total		—	$ 105.05	—	—	$ 0.76	—
Net profit		—	—	—	—	$ 104.29	—

Note: Sources for average meat price/kg is from [119] and source to determine the price of biochar/kg is from [120].

Biochar can affect methane production in the rumen. Methanogenesis is a microbiological activity that produces methane as a by-product. Reducing circumstances, helped by biochar's and redox characteristics, influence the activity of methane-producing microorganisms which redox-active chemicals in biochar function as electron acceptors, altering microbial processes and possibly lowering methane emissions [118]. Similarly, Abdel-Tawwab, El-Sayed, and Shady [42] noted that the sored of nitrate in the particles of biochar, which is created by NO^{-}_{3} reduction in the rumen, cannot enter the bloodstream and prevented is a major factor in lowering CH_4 generation during rumen digestion. The ruminal microbiota would be directly impacted by NO^{-}_{2}, which would inhibit methanogenesis [42].

10. Cost-Effectiveness of Using Biochar as a Feed Additive on a Large-Scale Livestock Production System

As presented in Table 10, there is a significant profit gain per animal product (kg of meat) when biochar is used as a feed additive. Adding biochar to a feed is a low-cost effective method of boosting livestock production margins. Biochar enhances feed efficiency and animals' well-being. It can significantly increase herd longevity and lowering replacement costs [121]. As average body weight gain results extracted from Table 3 revealed that, it is possible to gain about $ 104.29 per all listed animals. Therefore, we can conclude that by using the correct level of inclusion, it is possible to improve the production cost of livestock and gain a significant profit.

11. Long-Term Feeding and Its Impact on Animal Performance, Gut Micro Flora, and Product Quality

In this review, we did not find a serious side effect of biochar used as a feeding additive for farm animals. There are few reports about the negative effect of biochar as a feed additive. Without considering the biochar's preparation, using it as a feed additive may hazard rather than increasing growth performance and animal products like egg, meat, and milk [3]. According to Schmidt et al., [17] activated biochar may sometimes adsorb useful and healthy bacterial flora. They stated that long-term biochar feeding may shift microbiota dynamics in the digestion system of animals, and there must be more methodological investigation for digestive and harmful microorganisms before reaching an overall conclusion. In addition, a few negative effects were identified regarding the immobilization of liposoluble feed ingredients (e.g., vitamin E or Carotenoids), which may limit long-term biochar feeding [17]. Additionally, Fujita et al. [122] stated that the inclusion of 0.5% biochar in daily fed of hens affects vitamin E content in the eggs by 40%. Animals fed with biochar have shown improvements in meat quality, including better fat composition, reduced cholesterol levels, and enhanced flavor. In addition, biochar supplementation can lead to lower levels of harmful residues in meat, potentially extending shelf life and improving safety. In dairy cows and laying hens, biochar supplementation has been linked to higher nutrient content in milk and eggs, including increased levels of beneficial fatty acids and vitamins [3, 17].

12. Conclusion and Recommendation

In this review, we can conclude that adding biochar to an animal's diet has an important effect on the animal's growth, ADWG, FCR, digestibility, microbiome dynamics, and ability to reduce intestinal methane output. Feeding biochar used as a feed additive has a positive effect on animal body weight gain and reduces methane (CH_4) emission, it means that we can improve the feeding system of ruminants to mitigate methane emissions, and an environmental management system is possible. This also results in animals' body weight gain. For example, as the results inTable 1 reveal, we can include up to 8%, 6%, 3%, 1%, and 1.5% of biochar to get maximum weight gain for fattened bull 2 kg, dairy cow 2.3 kg, broiler 0.2 kg, pig 4 kg, and sheep 3.1 kg, respectively. Therefore, based on our review, we would like to recommend the following points and future considerations.

- Using biochar as a feed additive in livestock production is very important since it increases body weight gain, feed conversion efficiency, and digestibility of animals.
- Environmental pollution can be improved by the mitigation of greenhouse gases from ruminants via biochar application as a feed additive.
- Further research is needed to determine the standard and maximum level of biochar inclusion.

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- More detailed bibliometric data analysis on the use of biochar is important to address biochar applications for all published research fields.

Data Availability Statement

This review is conducted based on published articles from different databases like Scopus-indexed journals, PubMed, and other search engines.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Conceptualization and original draft: Bernabas Ayeneshet and Tenaw Temesgen

Source Collection, Writing, review AND editing: All authors collected the relevant published article from highly indexed journals, wrote the full review article and contributed to the review article, and approved the submitted version.

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References

[1] B. M. Naveena and M. Kiran, "Buffalo Meat Quality, Composition, and Processing Characteristics: Contribution to the Global Economy and Nutritional Security," *Animal Frontiers* 4, no. 4 (2014): 18–24.

[2] A. Asresie, L. Zemedu, and E. Adigrat, "The Contribution of Livestock Sector in Ethiopian Economy," *Review Advances in Life Science* 29 (2015).

[3] K. Y. Man, K. L. Chow, Y. B. Man, W. Y. Mo, and M. H. Wong, "Use of Biochar as Feed Supplements for Animal Farming," *Critical Reviews in Environmental Science and Technology* 51, no. 2 (2021): 187–217.

[4] H. Abebaye, A. Mengistu, B. Tamir, G. Assefa, and F. Feyissa, "Feed Resources Availability and Feeding Practices of Smallholder Farmers in Selected Districts of West Shewa Zone, Ethiopia," *World Journal of Agricultural Sciences* 15, no. 1 (2019): 21–30.

[5] U. Okereafor, M. Makhatha, L. Mekuto, N. Uche-Okereafor, T. Sebola, and V. Mavumengwana, "Toxic Metal Implications on Agricultural Soils, Plants, Animals, Aquatic Life and Human Health," *International Journal of Environmental Research* 17, no. 7 (2020): 2204.

[6] M. Zafar-ul-Hye, M. Tahzeeb-ul-Hassan, M. Abid, et al., "Potential Role of Compost Mixed Biochar With Rhizobacteria in Mitigating Lead Toxicity in Spinach," *Scientific Reports* 10, no. 1 (2020): 12159.

[7] A. A. Rahi, U. Younis, N. Ahmed, et al., "Toxicity of Cadmium and Nickel in the Context of Applied Activated Carbon Biochar for Improvement in Soil Fertility," *Saudi Journal of Biological Sciences* 29, no. 2 (2022): 743–750.

[8] K. N. Palansooriya, J. Li, P. D. Dissanayake, et al., "Prediction of Soil Heavy Metal Immobilization by Biochar Using Machine Learning," *Environmental Science & Technology* 56, no. 7 (2022): 4187–4198.

[9] S. Acheampong, "Heavy Metals' Poisoning in Farm Animals," in *Heavy Metals-Recent Advances*, (IntechOpen, 2023).

[10] K. Kalus, J. A. Koziel, and S. Opaliński, "A Review of Biochar Properties and Their Utilization in Crop Agriculture and Livestock Production," *Applied Sciences* 9, no. 17 (2019): 3494.

[11] K. Abhishek, A. Shrivastava, V. Vimal, et al., "Biochar Application for Greenhouse Gas Mitigation, Contaminants Immobilization and Soil Fertility Enhancement: A State-of-the-Art Review," *Science of the Total Environment* 853 (2022): 158562.

[12] M. O. Akca and O. Sonmez, "Role of Biochar in the Adsorption of Heavy Metals," in *Sustainable Agriculture Reviews 61: Biochar to Improve Crop Production and Decrease Plant Stress under a Changing Climate*, (Springer, 2023): 293–307.

[13] B. Glaser, "Prehistorically Modified Soils of Central Amazonia: A Model for Sustainable Agriculture in the Twenty-First Century," *Philosophical Transactions of the Royal Society B: Biological Sciences* 362, no. 1478 (2007): 187–196.

[14] J. Lehmann and S. Joseph, *Biochar for Environmental Management: Science, Technology and Implementation* (Routledge, 2015).

[15] N. Rashidi, A. Khatibjoo, K. Taherpour, M. Akbari-Gharaei, and H. Shirzadi, "Effects of Licorice Extract, Probiotic, Toxin Binder and Poultry Litter Biochar on Performance, Immune Function, Blood Indices and Liver Histopathology of Broilers Exposed to Aflatoxin-B1," *Poultry Science* 99, no. 11 (2020): 5896–5906.

[16] G. Wu, "Management of Metabolic Disorders (Including Metabolic Diseases) in Ruminant and Nonruminant Animals," in *Animal Agriculture*, (Elsevier, 2020): 471–491.

[17] H.-P. Schmidt, N. Hagemann, K. Draper, and C. Kammann, "The Use of Biochar in Animal Feeding," *PeerJ* 7 (2019): e7373.

[18] S. Tang, Z. Wang, Z. Liu, Y. Zhang, and B. Si, "The Role of Biochar to Enhance Anaerobic Digestion: A Review," *Journal of Renewable Materials* 8, no. 9 (2020): 1033–1052.

[19] B. Group, "How Biochar Can Reduce Methane Emissions from Animal Husbandry," 2023, [cited 2024 12/05/2024]; Available from: https://www.environmental-expert.com/articles/how-biochar-can-reduce-methane-emissions-from-animal-husbandry-1111722.

[20] M. Islam, S. Ahmed, Y. Kim, H. Mun, and C. Yang, "Effect of Sea Tangle (*Laminaria japonica*) and Charcoal Supplementation as Alternatives to Antibiotics on Growth Performance and Meat Quality of Ducks," *Asian-Australasian Journal of Animal Sciences* 27, no. 2 (2014): 217–224.

[21] C. Kammann, J. Ippolito, N. Hagemann, et al., "Biochar as a Tool to Reduce the Agricultural Greenhouse-Gas Burden-Knowns, Unknowns and Future Research Needs," *Journal of*

9403, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/aia/9911760 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Environmental Engineering Landscape Management 25, no. 2 (2017): 114–139.

[22] M. Eger, M. Graz, S. Riede, and G. Breves, "Application of MootralTM Reduces Methane Production by Altering the Archaea Community in the Rumen Simulation Technique," *Frontiers in Microbiology* 9 (2018): 2094.

[23] S. A. Terry, G. O. Ribeiro, R. J. Gruninger, et al., "A Pine Enhanced Biochar Does Not Decrease Enteric CH_4 Emissions, But Alters the Rumen Microbiota," *Frontiers in Veterinary Science* 6 (2019): 308.

[24] G. M. Chu, C. K. Jung, H. Y. Kim, et al., "Effects of Bamboo Charcoal and Bamboo Vinegar as Antibiotic Alternatives on Growth Performance, Immune Responses and Fecal Microflora Population in Fattening Pigs," *Animal Science Journal* 84, no. 2 (2013): 113–120.

[25] T. P. Prasai, K. B. Walsh, S. P. Bhattarai, et al., "Biochar, Bentonite and Zeolite Supplemented Feeding of Layer Chickens Alters Intestinal Microbiota and Reduces Campylobacter Load," *PLOS ONE* 11, no. 4 (2016): e0154061.

[26] R. Teoh, E. Caro, D. B. Holman, S. Joseph, S. J. Meale, and A. V. Chaves, "Effects of Hardwood Biochar on Methane Production, Fermentation Characteristics, and the Rumen Microbiota Using Rumen Simulation," *Frontiers in Microbiology* 10 (2019): 1534.

[27] P. J. Tamayao, G. O. Ribeiro, T. A. McAllister, et al., "Effects of Post-Pyrolysis Treated Biochars on Methane Production, Ruminal Fermentation, and Rumen Microbiota of a Silage-Based Diet in an Artificial Rumen System (RUSITEC)," *Animal Feed Science and Technology* 273 (2021): 114802.

[28] M. H. Sirjani, J. R. M. Zahedifar, and Y. Rouzbehan, "Effect of Adding Biochar in Diets Containing Probiotics on in Vitro Fermentation Variables, Health Indicators, Rectum Bacteria, and Blood Enzymes of Holstein Calves," *Animal Production Research* 11 (2022): 1–19.

[29] A. Kumar, T. Bhattacharya, W. A. Shaikh, et al., "Multifaceted Applications of Biochar in Environmental Management: A Bibliometric Profile," *Biochar* 5, no. 1 (2023): 11.

[30] H. P. Schmidt, T. Bucheli, C. Kammann, B. Glaser, S. Abiven, and J. Leifeld, "European Biochar Certificate-Guidelines for a Sustainable Production of Biochar," (2016).

[31] A. M. Evans, J. W. Boney, and J. S. Moritz, "The Effect of Poultry Litter Biochar on Pellet Quality, One to 21 d Broiler Performance, Digesta Viscosity, Bone Mineralization, and Apparent Ileal Amino Acid Digestibility," *Journal of Applied Poultry Research* 26, no. 1 (2017): 89–98.

[32] B. Sivilai and T. Preston, "Rice Distillers' Byproduct and Biochar as Additives to a Forage-Based Diet for Native Moo Lath Sows During Pregnancy and Lactation," *Livestock Research for Rural Development* 31, no. 10 (2019).

[33] R. Leng, T. Preston, and S. Inthapanya, "Biochar Reduces Enteric Methane and Improves Growth and Feed Conversion in Local "Yellow" Cattle Fed Cassava Root Chips and Fresh Cassava Foliage," *Livestock Research for Rural Development* 24, no. 11 (2012): 1–7.

[34] T. Le Thi Thuy Hang, N. X. Ba, and D. Van Dung, "Effect of Biochar on Growth and Methane Emissions of Goats Fed Fresh Cassava Foliage," *Livestock Research for Rural Development* 31 (2019): 5.

[35] P. Silivong and T. Preston, "Growth Performance of Goats Was Improved When a Basal Diet of Foliage of *Bauhinia acuminata* Was Supplemented With Water Spinach and Biochar," *Livestock Research for Rural Development* 27, no. 3 (2015): 58.

[36] D. Keba, T. Tolemariam, S. Demeke, and A. Alkhtib, "Corncob Biochar Supplementation Improves Nutrient Digestibility, Fattening Performance and Carcass Characteristics of Fattening Sheep," *Veterinary Medicine and Science* 9, no. 2 (2023): 967–973.

[37] C. Dim, E. Akuru, M. Egom, et al., "Effect of Dietary Inclusion of Biochar on Growth Performance, Haematology and Serum Lipid Profile of Broiler Birds," *Agro-Science* 17, no. 2 (2018): 9–17.

[38] R. Kleyn, *Chicken Nutrition: A Guide for Nutritionists and Poultry Professionals* (Context, 2013).

[39] T. T. Lan, T. Preston, and R. Leng, "Feeding Biochar or Charcoal Increased the Growth Rate of Striped Catfish (*Pangasius hypophthalmus*) and Improved Water Quality," *Livestock Research for Rural Development* 28, no. 5 (2016): 84.

[40] N. D. Khaki, "Innovative Approaches in Aquaculture for Conservation and Management of Threatened Fish Species: The Case of Different Strains of Brown Trout (*Salmo trutta* and *Salmo ghigii*)," (2018).

[41] J. W. MacAdam and J. Villalba, "Beneficial Effects of Temperate Forage Legumes that Contain Condensed Tannins," *Agriculture* 5, no. 3 (2015): 475–491.

[42] M. Abdel-Tawwab, G. O. El-Sayed, and S. H. Shady, "Effect of Dietary Active Charcoal Supplementation on Growth Performance, Biochemical and Antioxidant Responses, and Resistance of Nile Tilapia, *Oreochromis niloticus* (L.) to Environmental Heavy Metals Exposure," *Aquaculture* 479 (2017): 17–24.

[43] R. Leng, "Interactions Between Microbial Consortia in Biofilms: A Paradigm Shift in Rumen Microbial Ecology and Enteric Methane Mitigation," *Animal Production Science* 54, no. 5 (2014): 519–543.

[44] T. M. Winders, M. L. Jolly-Breithaupt, H. C. Wilson, J. C. MacDonald, G. E. Erickson, and A. K. Watson, "Evaluation of the Effects of Biochar on Diet Digestibility and Methane Production From Growing and Finishing Steers," *Translational Animal Science* 3, no. 2 (2019): 775–783.

[45] A. Mirheidari, N. M. Torbatinejad, P. Shakeri, and A. Mokhtarpour, "Effects of Biochar Produced From Different Biomass Sources on Digestibility, Ruminal Fermentation, Microbial Protein Synthesis and Growth Performance of Male Lambs," *Small Ruminant Research* 183 (2020): 106042.

[46] K. Liu, Y. Zhang, Z. Yu, et al., "Ruminal Microbiota-Host Interaction and Its Effect on Nutrient Metabolism," *Animal Nutrition* 7, no. 1 (2021): 49–55.

[47] S. Yu, W. Shi, B. Yang, et al., "Effects of Repeated Oral Inoculation of Artificially Fed Lambs With Lyophilized Rumen Fluid on Growth Performance, Rumen Fermentation, Microbial Population and Organ Development," *Animal Feed Science and Technology* 264 (2020): 114465.

[48] L. L. Guan, J. D. Nkrumah, J. A. Basarab, and S. S. Moore, "Linkage of Microbial Ecology to Phenotype: Correlation of Rumen Microbial Ecology to Cattle's Feed Efficiency," *FEMS Microbiology Letters* 288, no. 1 (2008): 85–91.

[49] M. Amean and T. Shujaa, "Effect of Adding Biochar to Ration on Productive Performance and Some Carcass Characteristics of Iraqi Awassi Lambs," *Tikrit Journal for Agricultural Sciences* 19, no. 4 (2019): 16–22.

[50] B. Bowman, "Rumen Microbes the New Hot Topic of Cattle Nutrition Research," (2019).

[51] H. A. Paz, C. L. Anderson, M. J. Muller, P. J. Kononoff, and S. C. Fernando, "Rumen Bacterial Community Composition

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in Holstein and Jersey Cows is Different Under Same Dietary Condition and is Not Affected by Sampling Method," *Frontiers in Microbiology* 7 (2016): 1206.

[52] Z. Zhu, S. J. Noel, G. F. Difford, et al., "Community Structure of the Metabolically Active Rumen Bacterial and Archaeal Communities of Dairy Cows Over the Transition Period," *PLOS ONE* 12, no. 11 (2017): e0187858.

[53] H. Thoetkiattikul, W. Mhuantong, T. Laothanachareon, et al., "Comparative Analysis of Microbial Profiles in Cow Rumen Fed With Different Dietary Fiber by Tagged 16S rRNA Gene Pyrosequencing," *Current Microbiology* 67, no. 2 (2013): 130–137.

[54] R. W. Li, C. D. Moon, and D. P. Morgavi, "Rumen Microbiome Dynamics and Their Implications in Health and Environment," *Frontiers in Microbiology* 14 (2023): 1223885.

[55] H. Mao, Y. Zhang, Y. Yun, et al., "Weaning Age Affects the Development of the Ruminal Bacterial and Archaeal Community in Hu Lambs During Early Life," *Frontiers in Microbiology* 12 (2021): 636865.

[56] Y. Sha, X. Liu, X. Pu, et al., "Characterizing the Dynamics of the Rumen Microbiota, Its Metabolites, and Blood Metabolites Across Reproductive Stages in Small-Tailed Han Sheep," *Microbiology Spectrum* 11, no. 6 (2023): e02867–e02823.

[57] M. Rey, F. Enjalbert, S. Combes, L. Cauquil, O. Bouchez, and V. Monteils, "Establishment of Ruminal Bacterial Community in Dairy Calves From Birth to Weaning is Sequential," *Journal of Applied Microbiology* 116, no. 2 (2014): 245–257.

[58] D. T. Eberl, M. J. Smith, O. J. Megram, et al., "Innovative Bedding Materials for Compost Bedded Pack Barns: Enhancing Dairy Cow Welfare and Sustainable Dairy Farming," *Environment, Development and Sustainability* (2024): 1–25.

[59] G. Agegnehu, A. K. Srivastava, and M. I. Bird, "The Role of Biochar and Biochar-Compost in Improving Soil Quality and Crop Performance: A Review," *Applied Soil Ecology* 119 (2017): 156–170.

[60] A. Salma, L. Fryda, and H. Djelal, "Biochar: A Key Player in Carbon Credits and Climate Mitigation," *Resources* 13, no. 2 (2024): 31.

[61] S. A. Razzak, "Municipal Solid and Plastic Waste Derived High-Performance Biochar Production: A Comprehensive Review," *Journal of Analytical and Applied Pyrolysis* 181 (2024): 106622.

[62] N. Lodhi, S. N. Chadar, D. S. Thakur, and A. Raikwar, "A Comprehensive Study on Biochar-Based Nanocomposites in Removal of Organic Pollutants From Wastewater," *Journal of Water and Environmental Nanotechnology* 9, no. 3 (2024): 302–317.

[63] G. De Bhowmick, A. K. Sarmah, and R. Sen, "Zero-Waste Algal Biorefinery for Bioenergy and Biochar: A Green Leap Towards Achieving Energy and Environmental Sustainability," *Science of the Total Environment* 650 (2019): 2467–2482.

[64] O. Zaid, F. Alsharari, and M. Ahmed, "Utilization of Engineered Biochar as a Binder in Carbon Negative Cement-Based Composites: A Review," *Construction and Building Materials* 417 (2024): 135246.

[65] T. Whitman, S. M. Scholz, and J. Lehmann, "Biochar Projects for Mitigating Climate Change: An Investigation of Critical Methodology Issues for Carbon Accounting," *Carbon Management* 1, no. 1 (2014): 89–107.

[66] Z. Dai, X. Xiong, H. Zhu, et al., "Association of Biochar Properties with Changes in Soil Bacterial, Fungal and Fauna Communities and Nutrient Cycling Processes," *Biochar* 3, no. 3 (2021): 239–254.

[67] M. Bartoli, R. Arrigo, G. Malucelli, A. Tagliaferro, and D. Duraccio, "Recent Advances in Biochar Polymer Composites," *Polymers* 14, no. 12 (2022): 2506.

[68] Y. Zhou, Z. Zhang, and M. K. Awasthi, "Exploring the Impact of Biochar Supplement on the Dynamics of Antibiotic Resistant Fungi During Pig Manure Composting," *Environmental Pollution* 314 (2022): 120235.

[69] S. Phongphanith and T. Preston, "Effect of Rice-Wine Distillers' Byproduct and Biochar on Growth Performance and Methane Emissions in Local "Yellow" Cattle Fed Ensiled Cassava Root, Urea, Cassava Foliage and Rice Straw," 28 (2018).

[70] J. Sperber, B. Troyer, G. E. Erickson, and A. K. Watson, "Evaluation of the Effects of Pine-Sourced Biochar on Cattle Performance and Methane and Carbon Dioxide Production From Growing and Finishing Steers," *Translational Animal Science* 6, no. 4 (2022): txac152.

[71] K. Kalus, D. Konkol, M. Korczyński, J. A. Koziel, and S. Opaliński, "Effect of Biochar Diet Supplementation on Chicken Broilers Performance, NH_3 and Odor Emissions and Meat Consumer Acceptance," *Animals* 10, no. 9 (2020): 1539.

[72] O. I. A. Al-Jumaily and T. K. H. Al-Jumaily, "Effect of Biochar on Growth Performance, Gas Concentration and Broiler Leg Health of Broilers," *Tikrit Journal for Agricultural Sciences* 22, no. 4 (2022): 7–15.

[73] N. N. Hien, N. N. X. Dung, L. H. Manh, and B. T. L. Minh, "Effects of Biochar Inclusion in Feed and Chicken Litter on Growth Performance, Plasma Lipids and Fecal Bacteria Count of Noi Lai Chicken," (2018).

[74] M. A. Khalid, S. M. Hussain, S. Mahboob, K. A. Al-Ghanim, and M. N. Riaz, "Biochar as a Feed Supplement for Nutrient Digestibility and Growth Performance of *Catla catla* Fingerlings," *Saudi Journal of Biological Sciences* 29, no. 12 (2022): 103453.

[75] B. Sivilai and T. Preston, "Rice Distillers' Byproduct and Biochar as Additives to a Forage-Based Diet for Native Moo Lath Sows During Pregnancy and Lactation," *Livestock Research for Rural Development* 31 (2019): 151.

[76] P. Silivong and T. Preston, "Supplements of Water Spinach (*Ipomoea aquatica*) and Biochar Improved Feed Intake, Digestibility, N Retention and Growth Performance of Goats Fed Foliage of *Bauhinia acuminata* as the Basal Diet," *Livestock Research for Rural Development* 28, no. 5 (2016).

[77] D. J. McAvoy, B. Burritt, and J. J. Villalba, "Use of Biochar by Sheep: Impacts on Diet Selection, Digestibility, and Performance," *Journal of Animal Science* 98, no. 12 (2020): skaa380.

[78] G. Bonanomi, M. Idbella, and A. M. Abd-ElGawad, "Microbiota Management for Effective Disease Suppression: A Systematic Comparison Between Soil and Mammals Gut," *Sustainability* 13, no. 14 (2021): 7608.

[79] J. Hou, A. Pugazhendhi, T. N. Phuong, et al., "Plant Resistance to Disease: Using Biochar to Inhibit Harmful Microbes and Absorb Nutrients," *Environmental Research* 214 (2022): 113883.

[80] D. D. Warnock, J. Lehmann, T. W. Kuyper, and M. C. Rillig, "Mycorrhizal Responses to Biochar in Soil-Concepts and Mechanisms," *Plant and Soil* 300 (2007): 9–20.

[81] Y. Li, C. Cheng, H. Wang, et al., "Distribution, Toxicity, and Impacts of Nano-Biochar in Mice Following Dietary Exposure: Insights Into Environmental Risks and Mammalian Effects," *Environmental Pollution* 338 (2023): 122652.

9403, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/aia/9911760 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[82] X. Yang, R. Hou, Q. Fu, et al., "A Critical Review of Biochar as an Environmental Functional Material in Soil Ecosystems for Migration and Transformation Mechanisms and Ecological Risk Assessment," *Journal of Environmental Management* 360 (2024): 121196.

[83] L. Xiang, J. D. Harindintwali, F. Wang, et al., "Integrating Biochar, Bacteria, and Plants for Sustainable Remediation of soils Contaminated With Organic Pollutants," *Environmental Science & Technology* 56, no. 23 (2022): 16546–16566.

[84] C. A. Masiello, Y. Chen, X. Gao, et al., "Biochar and Microbial Signaling: Production Conditions Determine Effects on Microbial Communication," *Environmental Science & Technology* 47, no. 20 (2013): 11496–11503.

[85] R. Leng, S. Inthapanya, and T. Preston, "All Biochars Are Not Equal in Lowering Methane Production in in Vitro Rumen Incubations," *Livestock Research for Rural Develoment* 12 (2013): 12.

[86] T. M. Winders, C. B. Freeman, B. A. Mark, et al., "Biochar Supplementation in Growing and Finishing Diets," (2019).

[87] A. M. Saleem, G. O. Ribeiro Jr, W. Z. Yang, et al., "Effect of Engineered Biocarbon on Rumen Fermentation, Microbial Protein Synthesis, and Methane Production in an Artificial Rumen (RUSITEC) Fed a High Forage Diet," *Journal of Animal Science* 96, no. 8 (2018): 3121–3130.

[88] P. S. Erickson, N. L. Whitehouse, and M. L. Dunn, "Activated Carbon Supplementation of Dairy Cow Diets: Effects on Apparent Total-Tract Nutrient Digestibility and Taste Preference," *The Professional Animal Scientist* 27, no. 5 (2011): 428–434.

[89] H. H. Hansen, I. M. L. D. Storm, and A. M. Sell, "Effect of Biochar on in Vitro Rumen Methane Production," *Acta Agriculturae Scandinavica, Section A—Animal Science* 62, no. 4 (2012): 305–309.

[90] Z. D. McFarlane, P. R. Myer, E. R. Cope, et al., "Effect of Biochar Type and Size on in Vitro Rumen Fermentation of Orchard Grass Hay," (2017).

[91] M. Al-Azzawi, L. Bowtell, K. Hancock, and S. Preston, "Addition of Activated Carbon into a Cattle Diet to Mitigate GHG Emissions and Improve Production," *Sustainability* 13, no. 15 (2021): 8254.

[92] G. Terler, M. Winter, M. Mandl, J. Sweeney, and A. Steinwidder, "Effect of Biochar or Biochar and Urea Supplementation on Feed Intake, Milk Yield, Feed Conversion and Methane Production of Dairy Cows," *Czech Journal of Animal Science* 68, no. 6 (2023): 245–254.

[93] R. C. Pereira, S. Muetzel, M. C. Arbestain, et al., "Assessment of the Influence of Biochar on Rumen and Silage Fermentation: A Laboratory-Scale Experiment," *Animal Feed Science Technology* 196 (2014): 22–31.

[94] A. Clark, *Managing Cover Crops Profitably* (Diane Publishing, 2008).

[95] I. Cabeza, T. Waterhouse, S. Sohi, and J. A. Rooke, "Effect of Biochar Produced From Different Biomass Sources and at Different Process Temperatures on Methane Production and Ammonia Concentrations in Vitro," *Animal Feed Science and Technology* 237 (2018): 1–7.

[96] J. L. Black, T. M. Davison, and I. Box, "Methane Emissions From Ruminants in Australia: Mitigation Potential and Applicability of Mitigation Strategies," *Animals* 11, no. 4 (2021): 951.

[97] V. Sejian and S. M. K. Naqvi, "Livestock and Climate Change: Mitigation Strategies to Reduce Methane Production," *Greenhouse Gases-Capturing, Utilization and Reduction* (2012): 255–276.

[98] S. Jeffery, F. G. Verheijen, C. Kammann, and D. Abalos, "Biochar Effects on Methane Emissions From Soils: A Meta-Analysis," *Soil Biology Biochemistry* 101 (2016): 251–258.

[99] R. Leng, S. Inthapanya, and T. Preston, "Biochar Lowers Net Methane Production From Rumen Fluid in Vitro," *Livestock Research for Rural Development* 24, no. 6 (2012): 103.

[100] L. Yu, Y. Yuan, J. Tang, Y. Wang, and S. Zhou, "Biochar as an Electron Shuttle for Reductive Dechlorination of Pentachlorophenol by *Geobacter sulfurreducens*," *Scientific Reports* 5, no. 1 (2015): 16221.

[101] A. F. Ramos, S. A. Terry, D. B. Holman, et al., "Tucumã Oil Shifted Ruminal Fermentation, Reducing Methane Production and Altering the Microbiome but Decreased Substrate Digestibility Within a RUSITEC Fed a Mixed Hay-Concentrate Diet," *Frontiers in Microbiology* 9 (2018): 1647.

[102] S. A. Terry, A. F. Ramos, D. B. Holman, T. A. McAllister, G. Breves, and A. V. Chaves, "Humic Substances Alter Ammonia Production and the Microbial Populations Within a RUSITEC Fed a Mixed Hay-Concentrate Diet," *Frontiers in Microbiology* 9 (2018): 1410.

[103] A. C. Duarte, D. B. Holman, T. W. Alexander, Z. Durmic, P. E. Vercoe, and A. V. Chaves, "The Type of Forage Substrate Preparation Included as Substrate in a RUSITEC System Affects the Ruminal Microbiota and Fermentation Characteristics," *Frontiers in Microbiology* 8 (2017): 704.

[104] M. N. Haque, "Dietary Manipulation: A Sustainable Way to Mitigate Methane Emissions From Ruminants," *Journal of Animal Science* 60, no. 1 (2018): 1–10.

[105] V. Lind, A. Schwarm, M. Mele, et al., "Nutritional Approaches to Reduce Enteric Methane Emission From Ruminants," in *Technology for Environmentally Friendly Livestock Production*, (Springer, 2023): 65–98.

[106] E. Lao and E. Mbega, "Biochar as a Feed Additive for Improving the Performance of Farm Animals," (2020).

[107] I. Goiri, R. Ruiz, R. Atxaerandio, J. L. Lavin, X. D. de Otálora, and A. García-Rodríguez, "Assessing the Potential Use of a Feed Additive Based on Biochar on Broilers Feeding Upon Productive Performance, pH of Digestive Organs, Cecum Fermentation and Bacterial Community," *Animal Feed Science and Technology* 279 (2021): 115039.

[108] W. Ye, L. Li, Z. Tang, et al., "Biochar and Modified Magnetic Biochar Enhanced Anaerobic Digestion of Swine Wastewater Under Ammonia Stress: Performance and Microbial Dynamics," *Journal of Environmental Chemical Engineering* 12, no. 2 (2024): 111969.

[109] L. Verdi, A. Dalla Marta, S. Orlandini, A. Maienza, S. Baronti, and F. P. Vaccari, "Evaluation of Biochar Addition to Digestate, Slurry, and Manure for Mitigating Carbon Emissions," *Agriculture* 14, no. 1 (2024): 162.

[110] Z. Zhang, G. Huang, P. Zhang, J. Shen, S. Wang, and Y. Li, "Development of Iron-Based Biochar for Enhancing Nitrate Adsorption: Effects of Specific Surface Area, Electrostatic Force, and Functional Groups," *The Science of the Total Environment* 856, no. Pt 1 (2023): 159037.

[111] S.-L. Wu, W. Wei, Q. Xu, et al., "Revealing the Mechanism of Biochar Enhancing the Production of Medium Chain Fatty Acids From Waste Activated Sludge Alkaline Fermentation Liquor," *ACS ES&T Water* 1, no. 4 (2021): 1014–1024.

[112] Y. Yuan, N. Bolan, A. Prévoteau, et al., "Applications of Biochar in Redox-Mediated Reactions," *Bioresource Technology* 246 (2017): 271–281.

9403, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/aia/9911760 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[113] C. Yang, J. A. Rooke, I. Cabeza, and R. J. Wallace, "Nitrate and Inhibition of Ruminal Methanogenesis: Microbial Ecology, Obstacles, and Opportunities for Lowering Methane Emissions From Ruminant Livestock," *Frontiers in Microbiology* 7 (2016): 132.
[114] A. Kolganova, R. Lal, and J. Firkins, "Biochar's Electrochemical Properties Impact on Methanogenesis: Ruminal Vs Soil Processes," *Journal of Agricultural Chemistry Environment* 12, no. 1 (2023): 28–43.
[115] Z. Luo, B. Yao, X. Yang, et al., "Novel Insights Into the Adsorption of Organic Contaminants by Biochar: A Review," *Chemosphere* 287, no. Pt 2 (2022): 132113.
[116] L. Liang, F. Xi, W. Tan, X. Meng, B. Hu, and X. Wang, "Review of Organic and Inorganic Pollutants Removal by Biochar and Biochar-Based Composites," *Biochar* 3 (2021): 255–281.
[117] X.-F. Tan, S.-S. Zhu, R.-P. Wang, et al., "Role of Biochar Surface Characteristics in the Adsorption of Aromatic Compounds: Pore Structure and Functional Groups," *Chinese Chemical Letters* 32, no. 10 (2021): 2939–2946.
[118] H.-Y. Yuan, L.-J. Ding, E. F. Zama, P.-P. Liu, W. N. Hozzein, and Y.-G. Zhu, "Biochar Modulates Methanogenesis Through Electron Syntrophy of Microorganisms With Ethanol as a Substrate," *Environmental Science & Technology* 52, no. 21 (2018): 12198–12207.
[119] W. Bank, *Average Prices for Meat (beef) Worldwide From 2014 to 2025* (Statista, 2024).
[120] P. Zhang, W. Duan, H. Peng, B. Pan, and B. Xing, "Functional Biochar and Its Balanced Design," *ACS Environmental Au* 2, no. 2 (2022): 115–127.
[121] K. Wilson, "Costs and Returns of Biochar Use in Dairies," 2014, [cited 2024 05]; Available from: https://wilsonbiochar.com/.
[122] H. Fujita, K. Honda, R. Iwakiri, K. S. Guruge, N. Yamanaka, and N. Tanimura, "Suppressive Effect of Polychlorinated Dibenzo-p-Dioxins, Polychlorinated Dibenzofurans and Dioxin-Like Polychlorinated Biphenyls Transfer From Feed to Eggs of Laying Hens by Activated Carbon as Feed Additive," *Chemosphere* 88, no. 7 (2012): 820–827.
[123] S. Phongphanith and T. Preston, "Effect of Rice-Wine Distillers' Byproduct and Biochar on Growth Performance and Methane Emissions in Local "Yellow" Cattle Fed Ensiled Cassava Root, Urea, Cassava Foliage and Rice Straw," *Livestock Research for Rural Development* 28 (2018): 178.
[124] D. C. Schubert, B. Chuppava, F. Witte, N. Terjung, and C. Visscher, "Effect of Two Different Biochars as a Component of Compound Feed on Nutrient Digestibility and Performance Parameters in Growing Pigs," *Frontiers in Animal Science* 2 (2021): 633958.
[125] J. Pan, J. Ma, L. Zhai, and H. Liu, "Enhanced Methane Production and Syntrophic Connection Between Microorganisms During Semi-Continuous Anaerobic Digestion of Chicken Manure by Adding Biochar," *Journal of Cleaner Production* 240 (2019): 118178.

9403, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1155/aia/9911760 by National Health And Medical Research Council, Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License